Area 1: Basic Engineering Practice

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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1.A Engineering Terms, Symbols, and Drawings

1. Key concepts

- Orthographic views: US practice is third-angle projection. The top view sits above the front view and the right-side view sits to its right. First-angle (ISO-E) mirrors that layout.
- Section views show internal features. Full section: the cutting plane goes all the way through and the cut surface is hatched. Half section: one quarter is removed. Broken-out, revolved and removed sections are local variants.
- Dimension notation: a boxed dimension is BASIC (theoretically exact; the tolerance comes from a feature control frame). A dimension in parentheses is REFERENCE (information only, not inspected).
- GD&T (ASME Y14.5) has five families. Form (flatness, straightness, circularity, cylindricity) never takes a datum. Orientation (perpendicularity, parallelism, angularity), location (position, concentricity, symmetry) and runout (circular, total) need datums. Profile can go either way.
- Material conditions: MMC is the condition with the most material (smallest hole, largest shaft). LMC is the opposite. RFS is the default. A bonus tolerance applies when a feature departs from MMC and the MMC modifier is used.
- Virtual condition (VC) is the worst-case boundary. Shaft: VC = MMC + geometric tolerance. Hole: VC = MMC − geometric tolerance. VC sets the size of functional gauges and the mating clearance.

2. Essential formulas

Name	Formula	Variables / units
Bonus tolerance	$\text{Total position tol} = \text{stated tol} + \lvert\text{actual size} - \text{MMC size}\rvert$	valid only with the Ⓜ modifier; sizes in in or mm
Virtual condition (external feature)	$\text{VC} = \text{MMC} + \text{geometric tol}$	shaft/pin
Virtual condition (internal feature)	$\text{VC} = \text{MMC} - \text{geometric tol}$	hole
Position from coordinates	$\text{Diametral position error} = 2\sqrt{(\Delta x^2 + \Delta y^2)}$	Δx, Δy = measured deviation from the true position

3. Handbook search keywords

third angle projection · section view · GD&T · position tolerance · maximum material condition · virtual condition · datum

4. Worked examples

Example 1: Bonus tolerance and position check
A Ø0.500 +0.006/−0.000 in hole has position tolerance Ø0.010 at MMC. A hole measures Ø0.504, and its axis is 0.003 in off in x and 0.004 in off in y. Is it acceptable?

1. MMC of the hole = smallest size = 0.500 in.
2. Bonus = 0.504 − 0.500 = 0.004 in, so the allowed position = 0.010 + 0.004 = 0.014 in (diameter zone).
3. Actual diametral error = $2\sqrt{(0.003^2 + 0.004^2)} = 0.010$ in.
4. $0.010 \le 0.014$, so the hole is acceptable.

Answer: Acceptable (0.010 in used of 0.014 in allowed)

Example 2: Virtual condition of a pin
A pin is Ø0.750 ±0.002 in with perpendicularity Ø0.003 at MMC. Find its virtual condition.

1. MMC of the pin = largest size = 0.752 in.
2. VC = MMC + tol = 0.752 + 0.003 = 0.755 in.

Answer: 0.755 in

5. Common traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation $\sqrt{(\Delta x^2+\Delta y^2)}$ instead of the diametral $2\sqrt{(\Delta x^2+\Delta y^2)}$.
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).
- Mixing up first-angle and third-angle view placement.

1.A Engineering Terms, Symbols, and Drawings: one-page cheat summary

- ▶ 3rd angle: top view above, right view to the right
- ▶ Boxed = basic; (x) = reference
- ▶ Form tolerances take no datum
- ▶ Hole MMC = smallest; shaft MMC = largest
- ▶ Bonus = |actual − MMC| (Ⓜ only)
- ▶ Position error = $2\sqrt{(\Delta x^2+\Delta y^2)}$
- ▶ VC shaft = MMC + tol; VC hole = MMC − tol

Formulas

Bonus tolerance	Total position tol = stated tol + |actual size − MMC size|
Virtual condition (external feature)	VC = MMC + geometric tol
Virtual condition (internal feature)	VC = MMC − geometric tol
Position from coordinates	Diametral position error = $2\sqrt{(\Delta x^2 + \Delta y^2)}$

Search keywords

third angle projection, section view, GD&T, position tolerance, maximum material condition, virtual condition, datum

Top traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation $\sqrt{(\Delta x^2+\Delta y^2)}$ instead of the diametral $2\sqrt{(\Delta x^2+\Delta y^2)}$.
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).

1.B Project Management and Economic Analysis

1. Key concepts

- Time value of money: move every cash flow to the same point in time (P, F or A) at the interest rate i before you compare.
- Mutually exclusive alternatives: pick the best PW, AW or EUAC. With unequal lives, use annual worth, or PW over the least common multiple of the lives.
- Sunk costs are irrelevant. Only future cash flows that differ between alternatives count.
- Break-even: the volume at which revenue equals total cost. Payback ignores the time value of money and anything after payback.
- Depreciation: straight line D = (P − S)/n. MACRS uses table percentages. Book value = cost − accumulated depreciation.
- Project scheduling (CPM): the critical path is the LONGEST path through the network, and its activities have zero total float. Earned value: CPI = EV/AC, SPI = EV/PV (above 1 is good).

2. Essential formulas

Name	Formula	Variables / units
Single payment	$F = P(1+i)^{n}$; $P = F(1+i)^{-n}$	i per period, n periods
Uniform series PW	$P = A[(1+i)^{n}-1]/[i(1+i)^{n}]$	A at the end of each period
Capital recovery	$A = P{\cdot}i(1+i)^{n}/[(1+i)^{n}-1]$	(A/P,i,n)
Sinking fund	$A = F{\cdot}i/[(1+i)^{n}-1]$	(A/F,i,n)
EUAC	$\mathrm{EUAC} = P(A/P) - S(A/F) + \text{annual O\&M}$	S = salvage
Break-even	$N = \text{Fixed cost}/(\text{price} - \text{variable cost per unit})$	
Effective rate	i_eff = (1 + r/m)^m − 1	r nominal, m compoundings/yr
Earned value	$\mathrm{CPI} = \mathrm{EV}/\mathrm{AC}$; $\mathrm{SPI} = \mathrm{EV}/\mathrm{PV}$	

3. Handbook search keywords

interest factor tables · capital recovery · sinking fund · equivalent uniform annual cost · break-even · depreciation · critical path

4. Worked examples

Example 1: EUAC of a machine
A machine costs $50,000, lasts 10 years, has a $8,000 salvage value and $3,000/yr O&M. At i = 8%, find its EUAC.

1. (A/P,8%,10) = 0.14903; (A/F,8%,10) = 0.06903.
2. EUAC = 50,000(0.14903) − 8,000(0.06903) + 3,000 = $9,899/yr.

Answer: ≈ $9,899/yr

Example 2: Break-even volume
Fixed cost $90,000/yr, price $45/unit, variable cost $27/unit. Find the break-even volume.

1. N = 90,000/(45 − 27) = 5,000 units/yr.

Answer: 5,000 units/yr

5. Common traps

- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.
- Including sunk costs in a replacement study.
- Calling the path with the most activities the critical path (it is the longest in time).

1.B Project Management and Economic Analysis: one-page cheat summary

- ► $(A/P) = i(1+i)^n/[(1+i)^n-1]$
- ► $(A/F) = i/[(1+i)^n-1] = (A/P) - i$
- ► $(P/A) = 1/(A/P)$
- ► $EUAC = P(A/P) - S(A/F) + O\&M$
- ► Break-even $N = FC/(p - v)$
- ► SL depreciation $(P-S)/n$
- ► Critical path = longest, zero float

Formulas

Single payment	$F = P(1+i)^n$; $P = F(1+i)^{-n}$
Uniform series PW	$P = A[(1+i)^n-1]/[i(1+i)^n]$
Capital recovery	$A = P{\cdot}i(1+i)^n/[(1+i)^n-1]$
Sinking fund	$A = F{\cdot}i/[(1+i)^n-1]$
EUAC	EUAC = P(A/P) − S(A/F) + annual O&M
Break-even	N = Fixed cost/(price − variable cost per unit)
Effective rate	i_eff = (1 + r/m)^m − 1
Earned value	CPI = EV/AC ; SPI = EV/PV

Search keywords

interest factor tables, capital recovery, sinking fund, equivalent uniform annual cost, break-even, depreciation, critical path

Top traps

- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.

1.C Design Methodology

1. Key concepts

- The design process runs: define requirements → generate concepts → analyze and select (decision matrix) → detail design → prototype and test → release. It iterates.
- Verification asks 'did we build it right?' (does it meet the spec). Validation asks 'did we build the right thing?' (does it meet the user's need).
- FMEA rates each failure mode for Severity, Occurrence and Detection (1–10). RPN = S×O×D ranks the actions. A high severity alone also gets attention.
- Factor of safety n = strength/stress (or capacity/load). Reliability-based design uses distributions instead of a single n.
- Reliability: series R = ΠRᵢ; parallel (redundant) R = 1 − Π(1 − Rᵢ). Constant failure rate: R(t) = e^(−λt), MTBF = 1/λ.
- Root-cause failure analysis: preserve the evidence → fractography (beach marks show fatigue, a cup-cone shows ductile overload) → root cause → corrective action → verify.

2. Essential formulas

Name	Formula	Variables / units
Factor of safety	$n = S/\sigma$ (or allowable load/actual load)	
Series reliability	$R_s = R_1 R_2 \ldots R_n$	
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$	
Exponential reliability	$R(t) = e^{(-\lambda t)}$; MTBF $= 1/\lambda$	λ in failures/h
FMEA	$RPN = S \times O \times D$	each 1–10

3. Handbook search keywords

factor of safety · reliability series parallel · FMEA · failure rate · design process

4. Worked examples

Example 1: System reliability
Pump A (R = 0.95) feeds two redundant valves in parallel (each R = 0.90), then a controller (R = 0.98). Find the system reliability.

1. Valve pair: 1 − (0.10)² = 0.99.
2. System: 0.95 × 0.99 × 0.98 = 0.9217.

Answer: 0.922

Example 2: Exponential life
A component has λ = 2×10⁻⁵ failures/h. Find R at 10,000 h and the MTBF.

1. R = e^(−0.2) = 0.8187.
2. MTBF = 1/λ = 50,000 h.

Answer: R ≈ 0.819; MTBF = 50,000 h

5. Common traps

- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.C Design Methodology: one-page cheat summary

- ► Series: multiply R
- ► Parallel: $1 - \Pi(1-R)$
- ► $R = e^{(-\lambda t)}$, $MTBF = 1/\lambda$
- ► RPN = S·O·D
- ► n = strength/stress
- ► Verify = spec; validate = user need

Formulas

Factor of safety	$n = S/\sigma$ (or allowable load/actual load)
Series reliability	$R_s = R_1 R_2 \ldots R_n$
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$
Exponential reliability	$R(t) = e^{(-\lambda t)}$; $MTBF = 1/\lambda$
FMEA	$RPN = S \times O \times D$

Search keywords

factor of safety, reliability series parallel, FMEA, failure rate, design process

Top traps

- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.D Physical Properties of Materials

1. Key concepts

- Tension test: the elastic slope is E, then the 0.2% offset gives yield Sy, the peak gives the ultimate Su, then necking and fracture. Ductility is measured by %elongation or %reduction of area.
- True stress $\sigma_T = \sigma(1+\varepsilon)$ and true strain $\varepsilon_T = \ln(1+\varepsilon)$ hold up to necking (constant volume). Strain hardening: $\sigma_T = K\varepsilon_T^n$, and necking starts at $\varepsilon_T = n$.
- Resilience (elastic energy per volume) $u_r = Sy^2/2E$. Toughness is the area under the whole curve, approximately $(Sy+Su)/2 \cdot \varepsilon_f$.
- Hardness tracks strength for steels: Su ≈ 0.5·HB ksi (3.45·HB MPa).
- Fe–C diagram: eutectoid 0.76% C at 727 °C (pearlite). Hypoeutectoid steels hold proeutectoid ferrite plus pearlite; use the lever rule. Quenching gives martensite and tempering restores toughness.
- BCC metals (ferritic steels) have a ductile-to-brittle transition. FCC metals (Al, Cu, austenitic stainless) do not. Creep matters above about 0.4 T_m (absolute); the Larson–Miller parameter is LMP = T(C + log t).
- Thermal expansion: $\delta = \alpha L \Delta T$. Galvanic corrosion attacks the anode (e.g. steel against copper); avoid a small anode with a large cathode.
- Composites: the rule of mixtures $E_L = E_f V_f + E_m(1-V_f)$ applies along the fibers (iso-strain). Across the fibers (iso-stress): $1/E_T = V_f/E_f + V_m/E_m$.

2. Essential formulas

Name	Formula	Variables / units
Hooke	$\sigma = E\varepsilon$; $\varepsilon = \delta/L$	E: psi or GPa
Poisson	$\varepsilon_{lat} = -\nu \cdot \varepsilon_{axial}$	ν ≈ 0.3 for steel
True stress/strain	$\sigma_T = \sigma(1+\varepsilon)$; $\varepsilon_T = \ln(1+\varepsilon)$	before necking
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$	
Resilience	$u_r = Sy^2/(2E)$	in-lb/in³
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$	wt% C
Larson–Miller	$LMP = T(C + \log t)$	T absolute (°R or K), t in h, C ≈ 20
Thermal strain	$\delta = \alpha L \Delta T$	α per °F or °C
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$	

3. Handbook search keywords

stress-strain diagram · true stress · modulus of resilience · hardness conversion · iron-carbon phase diagram · lever rule · Larson-Miller · coefficient of thermal expansion · galvanic series

4. Worked examples

Example 1: Tension test reduction
A 0.500 in diameter steel bar ($E = 30\times10^6$ psi, $\nu = 0.29$) carries 12,000 lb. Find the stress, the axial strain and the change in diameter.

1. $A = \pi(0.5)^2/4 = 0.1963$ in²; $\sigma = 12{,}000/0.1963 = 61{,}115$ psi.
2. $\varepsilon = \sigma/E = 2.037\times10^{-3}$.
3. $\Delta d = -\nu \cdot \varepsilon \cdot d = -0.29 \times 2.037\times10^{-3} \times 0.5 = -2.95\times10^{-4}$ in (it contracts).

Answer: σ ≈ 61,115 psi; Δd ≈ 295 µin decrease

Example 2: Lever rule
A 0.30% C steel is slowly cooled to just below 727 °C (α at 0.022% C, eutectoid at 0.76% C). Find the fraction

of proeutectoid ferrite.

1. $W_{\alpha'} = (0.76 - 0.30)/(0.76 - 0.022) = 0.623$.
2. Pearlite = 0.377.

Answer: 0.623

Example 3: Larson–Miller
LMP = 38,000 (°R, h, C = 20). Find the rupture life at 1,200 °F.

1. T = 1,200 + 460 = 1,660 °R.
2. $\log t = 38{,}000/1{,}660 - 20 = 2.892 \rightarrow t = 779$ h.

Answer: ≈ 779 h

5. Common traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with $\sigma(1+\varepsilon)$ (the formula is invalid there).
- Putting the lever-rule numerator on the wrong side (use the OPPOSITE arm).
- Expecting FCC metals to show a DBTT.

1.D Physical Properties of Materials: one-page cheat summary

- ► $\sigma = P/A, \varepsilon = \delta/L, E = \sigma/\varepsilon$
- ► $\sigma_T = \sigma(1+\varepsilon), \varepsilon_T = \ln(1+\varepsilon)$
- ► $\%RA = 1 - (d_f/d_0)^2$
- ► $u_r = Sy^2/2E$
- ► $Su \approx 0.5\ HB\ ksi$
- ► Eutectoid 0.76% C, 727 °C
- ► $LMP = T_{abs}(20 + \log t)$
- ► $\delta = \alpha L \Delta T$
- ► $E_L = E_f V_f + E_m V_m$

Formulas

Hooke	$\sigma = E\varepsilon\ ;\ \varepsilon = \delta/L$
Poisson	$\varepsilon_{lat} = -\nu\varepsilon_{axial}$
True stress/strain	$\sigma_T = \sigma(1+\varepsilon);\ \varepsilon_T = \ln(1+\varepsilon)$
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$
Resilience	$u_r = Sy^2/(2E)$
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$
Larson–Miller	$LMP = T(C + \log t)$
Thermal strain	$\delta = \alpha L \Delta T$
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$

Search keywords

stress-strain diagram, true stress, modulus of resilience, hardness conversion, iron-carbon phase diagram, lever rule, Larson-Miller, coefficient of thermal expansion, galvanic series

Top traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with σ(1+ε) (the formula is invalid there).